

VESRION 3 SYSTEM



GrAid: A Hybrid Semantic and Vision-Based System for Automated Handwritten Exam Assessment

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CHAPTER 1 - THE PROBLEM AND ITS SETTINGS

Introduction

Educational assessment remains one of the most labor intensive and time-consuming processes in academic institutions. Although there have been rapid improvements in the digital learning technologies, manual checking is still critical in grading handwritten exams. Not only is this time consuming but it is also likely to experience inconsistent grading, teacher burnout and delayed academic feedback to students all of which can negatively affect both teaching quality and student learning outcomes. As class sizes grow and academic demands increase, these challenges highlight a pressing need for more efficient, scalable, and reliable assessment solutions.

The recent developments in Intelligent Document Processing (IDP) and artificial intelligence have made it possible to partially automate exam assessment, mainly on the objective question types like multiple-choice questions and short-answer questions. Conversely, essay type answers are also a major problem because of their subjectivism, the diversified handwriting, and the necessity of understanding of the semantics and not verbatim response. Current automated grading systems usually do not allow capturing subtlety, quality of reasoning, and expectations based on rubrics, which leads to the low rate of adoption of high-stakes academic assessments. These gaps highlight the need for systems that go beyond pure text extraction toward meaningful semantic interpretation of student responses.



Building on these trends and challenges, this study proposes the development of an AI-powered system designed to automatically check handwritten student examination papers, with a primary focus on essay-type responses. The proposed system integrates Surya for line detection and the Qwen2.5-VL-7B-Instruct vision-language model (4-bit quantized) for accurate transcription of handwritten essays, together with a semantic scoring mechanism using Groq (Llama 3) to evaluate student answers against instructor-defined rubrics. Rather than fully replacing human evaluators, the system adopts a Human-in-the-Loop approach, allowing teachers to review low-confidence or ambiguous cases to ensure fairness, transparency, and accountability in scoring. Rather than fully replacing human evaluators, the system adopts a Human-in-the-Loop approach, allowing teachers to review low-confidence or ambiguous cases to ensure fairness, transparency, and accountability in scoring.

The relevance of this research is on the possibility of modernizing the educational assessment practices where hand written examinations are still the norm. The system helps simplify the process of assessing essays by automating its evaluation, making it faster and the work of educators less demanding, and allowing it to be performed at scale. Ultimately, this study supports the transition toward intelligent, educator-centered assessment systems, allowing AI to handle scale without replacing educator judgment.

Several researchers have demonstrated the viability of AI-assisted grading in academic settings. Chaudhari (2025) showed that systems integrating OCR, NLP, and LLMs can digitize and semantically evaluate student responses at scale, while Himi et al. (2024) found that large



language models achieved error rates as low as 1.29% compared to human graders in zero-shot evaluation scenarios. Safilian et al. (2025) further established that rubric-aligned generative AI frameworks can produce not only numerical scores but also written justifications, addressing the transparency gap in automated assessment. These findings collectively reinforce the potential of a hybrid AI approach that combines physical document capture, handwriting recognition, and semantic evaluation, the core contributions of this study.

Background of the Study

Digitization of education has transformed the way we assess student performance, although there is a disparity between assessment of multiple choice questions which are easily automated and assessment of written work (Zhang, 2024). Even in the modern day classrooms, the process of grading essays and short answer questions is manual and tedious and in most cases it takes the teacher several days to complete the task. This high dependency on human labor causes inevitable issues in which now fatigued teachers fail to be consistent, and personal bias intrudes, which may hurt impartial evaluation and compromise credibility (Zhai et al., 2020). Optical Mark Recognition (OMR) is not new to the automated tests of multiple choices but it cannot yet take into consideration the subjective and qualitative results where students show their comprehension of the lesson in which they were taught. Although there has been progress in automated grading technology, the majority of the systems available today are designed to be used in entirely digital environments and provide no viable solutions to the situation of assessing handwritten essays with human validation in academies where resources are limited.



Although AI models with a high level of technology have been shown to be able to score as well as humans, they are generally based on an entirely digital setting, which most schools are yet to achieve (Zhang, 2024). Paper and handwriting is the reality in most areas and it leads to an inherent incompatibility as advanced digital technologies fail to reach physical student work. The key technical obstacles are the fact that the difference in the layout of documents is enormous, and it is not always possible to read the handwriting correctly when a photo is taken with a smartphone and has a bad light or even inconvenient angles. In addition to this, a great number of automated grading programs operate like black boxes, and they do not provide clear explanations in relation to actual rubrics, and hence they cannot be trusted or checked by a teacher.

This research study addresses these gaps through a five-stage hybrid grading platform anchored by standardized templates and a Human-in-the-Loop mechanism. First, professors use the built-in test-paper generator (fpdf2 + ArUco markers) to create printable A4 PDF templates containing DICT_4X4_50 ArUco corner markers and QR codes for exam/student identity; these templates organize question types into fixed, labeled zones with bubble sections for objective questions (planned extension) and bounded write-in boxes for identification and essay responses. Second, after examination, professors photograph the completed answer sheets using a standard smartphone and upload the images through a mobile-responsive React web dashboard, either individually or in batch. Third, the uploaded images are automatically preprocessed using OpenCV 4.10+ (grayscale conversion, deskewing, noise reduction, ArUco marker detection, and



perspective homography transform). Fourth, a dual-pipeline extraction system processes the image: an OMR pipeline (planned extension) uses OpenCV fill-ratio analysis while a hybrid OCR pipeline using Surya OCR 0.13.1 for line/region detection followed by Qwen2.5-VL-7B-Instruct (4-bit quantized via BitsAndBytes) extracts student responses. Fifth, the extracted responses are evaluated by a multi-method grading engine: exact-match logic scores multiple-choice and true/false answers, fuzzy string matching with partial credit handles identification items, and Groq (Llama-3.3-70B-Versatile) performs rubric-based semantic scoring for essays, generating both a numerical grade and a written justification.

Statement of the Problem

Educational assessment remains one of the most labor-intensive and time-consuming tasks for academic instructors. Despite advancements in educational technology, the evaluation of physical, handwritten examinations relies heavily on manual processing. This dependency often leads to delayed academic feedback, grading fatigue, and potential inconsistencies in scoring subjective responses. Existing automated systems are largely confined to objective question formats and are unable to process handwritten scripts or evaluate the semantic content of student essays. Furthermore, many contemporary AI scoring tools operate as opaque "black boxes," offering little transparency or manual override capabilities for the human educator.

This study seeks to answer the following questions:

1. How accurately can a hybrid OCR pipeline combining Surya OCR for line/region detection and Qwen2.5-VL-7B-Instruct (4-bit quantized) for handwriting recognition



convert raw handwritten student responses including variable or messy handwriting into digital text suitable for automated evaluation?

2. How can a hybrid grading engine applying exact-match logic for Multiple Choice and True/False questions, fuzzy string matching for Identification questions, and LLM-based semantic scoring via Groq (Llama-3.3-70B-Versatile) for essay responses automate examination checking across all four question types?
3. How can a confidence threshold mechanism and specialized dashboard be developed to flag low-confidence AI grades, keeping the human teacher in full control of the final score?
4. To what extent does the developed prototype reduce the perceived task load of General Education professors in checking papers compared to the traditional manual method, as measured by the NASA Task Load Index?

Objectives of the Study

The main objective of this study is to develop a practical hybrid AI-based system that assists teachers in checking handwritten essay examination papers by integrating pre-trained handwriting recognition, rubric-based semantic similarity scoring, and human validation to improve grading efficiency while maintaining fairness and teacher control.

Specifically, the study aims to:



- a) To develop a standardized answer sheet template and a mobile-responsive web platform for capturing and uploading physical exam papers using standard smartphone cameras.
- b) To integrate image preprocessing techniques, Surya line detection, and the Qwen2.5-VL-7B-Instruct vision-language model (4-bit) to reliably detect, crop, and transcribe handwritten responses from marker-guided template regions into digital text.
- c) To apply exact-match checking for objective questions and utilize Large Language Model inference via Groq (Llama-3.3-70B-Versatile) for rubric-based semantic essay evaluation.
- d) To create a confidence threshold mechanism that flags low-confidence AI transcriptions and provides a teacher dashboard for manual review, override, and score finalization.
- e) To evaluate the perceived task load reduction and user acceptability of the prototype among selected General Education professors.

Scope and Limitations of the Study

To achieve a functioning extraction and grading engine, the system utilizes a dual-engine pipeline. It employs Surya for line detection and Qwen2.5-VL-7B-Instruct (4-bit) for vision-based handwritten text recognition on cropped regions, alongside an Optical Mark Recognition (OMR) engine using OpenCV fill-ratio analysis for objective question types. The physical-to-digital interface is strictly scoped to standardized templates containing ArUco/QR markers, which enable automatic perspective warping and precise region cropping. Automated evaluation includes exact-match algorithms for objective questions and Groq (Llama-3.3-70B-Versatile) for rubric-based semantic essay scoring. The dashboard functions as a



decision-support tool, flagging low-confidence AI outputs for mandatory human review and final authority.

Operationally, the system is delimited to English handwriting, as the models are not optimized for multilingual scripts. Supported formats include JPG, JPEG, PNG, and WEBP with a 20MB maximum file size. While the system auto-up-scales images below 2,000 pixels, photos exceeding 8,000 pixels are discouraged due to increased processing latency.

The physical layout is further delimited by specific structural constraints to ensure accuracy:

- **Question Capacity:** Each test paper is capped at 4–5 essay questions per page (optimized at 3–4 questions) to maintain a minimum 25mm answer box for sufficient handwriting room.
- **Prompt Length:** Question prompts are recommended not to exceed 200 characters (~30–40 words) to ensure clarity and reduce student decoding time.
- **Input Quality:** Grading accuracy remains highly dependent on input quality; poor lighting, overlapping text, or excessive erasures will bypass the AI and require manual professor review.

Significance of the Study

With the continuous advancement of educational systems, basic and secondary schools require efficient and reliable assessment methods to manage growing workloads. This study is significant as it will use camera vision, image processing, optical character recognition (OCR) to



develop an automated test-paper checking system that will use AI-based evaluation to enhance grading efficiency, consistency and accuracy while minimizing manual checking.

This research would then be specifically significant and beneficial to the following:

To the Teachers. The proposed test-paper checking system allows teachers to focus more on academic planning and teaching as automating the evaluation process will reduce their workload associated with manual grading.

To the Students. The system promotes faster and consistent examination results and ensures fair evaluation and immediate feedback to the students' academic performance.

To the Schools and Educational Institutions. The system contributes by streamlining assessment procedures, supports efficient academic operations and reduces grading errors which is an efficient and cost-effective solution for managing examinations, particularly in large classes.

To the Researchers. This study provides the research team with hands-on experience in the practical application of artificial intelligence, computer vision, and software engineering within an educational context. It develops the researchers' competencies in system design, AI integration, empirical evaluation, and interdisciplinary problem-solving, contributing to their growth as Computer Science practitioners capable of designing technology-driven solutions to real-world challenges.



To Future Researchers and Developers. This study serves as a useful reference for future researchers in the development of camera vision and artificial intelligence-based assessment systems, encouraging further enhancements like improved handwriting recognition and adding question formats.

Overall, this study provides an accurate, efficient and practical automated test-paper checking system by demonstrating the applicability of camera vision and artificial-intelligence in enhancing assessment methods specifically for basic and secondary education.



CHAPTER 2 - REVIEW OF RELATED LITERATURE AND STUDIES

Overview

The landscape of Philippine higher education is currently characterized by a paradigm shift toward flexible learning, as mandated by Commission on Higher Education policies (Roa et al., 2023). However, handwritten examinations remain widely used, especially in high-stakes and general education courses, resulting in continued reliance on manual grading that, combined with large class sizes, increases workload and the risk of human error (Masagca, 2025; Roa et al., 2023).

The following sections review literature organized into five thematic areas: document digitization and information extraction, automated grading systems and AI applications, rubric-based and essay scoring frameworks, template-based standardization, and system architecture and implementation considerations.

The literature reviewed in this chapter examines key areas that form the foundation of the proposed system, including document digitization and information extraction, artificial intelligence-based grading mechanisms, rubric-based essay scoring frameworks, template standardization approaches, and system architecture considerations. Collectively, these studies provide both the theoretical grounding and practical insights necessary for designing a system that integrates mobile capture technology, OCR, AI-driven grading, and educator-centered web interfaces to address the persistent limitations of manual examination assessment. Recent research has established that the primary challenges in detecting messy handwriting stem from



its high intra- and inter-writer variability, stroke fragmentation, character overlap, inconsistent slant and sizing, and background noise in digitized documents. These characteristics create substantial obstacles for both human evaluators and automated optical character recognition (OCR) Kunhoth, J. & Al-Maadeed, et.al (2024).

Document Digitization and Information Extraction

The digitization of physical examination papers is a critical first stage in automated grading systems. The accuracy of text extraction directly influences the reliability of AI-based evaluation. Several studies emphasize that document structure plays a decisive role in successful information extraction.

- Alamoudi et al. (2021) developed a rule-based metadata extraction system utilizing Regular Expressions (Regex) and achieved an accuracy rate of 94.62%. Their findings suggest that structured document templates significantly enhance extraction precision while reducing computational requirements. However, they also identified layout variability as a major barrier to achieving perfect accuracy.
- Expanding beyond rule-based systems, Baviskar et al. (2021) examined the transition from traditional template-driven methods to AI-powered approaches involving Optical Character Recognition (OCR), Natural Language Processing (NLP), and Computer Vision (CV). While AI techniques improve flexibility in handling unstructured documents, the diversity of document layouts continues to challenge system robustness.



- Messy handwriting further complicates document digitization beyond layout variability. Garrido-Muñoz et al. (2026) noted that irregular strokes, inconsistent spacing, and varied writing styles increase variability, reducing OCR performance. Similarly, Ali and Abdulrazzaq (2023) emphasized that poor handwriting remains a major limitation in recognition systems, especially with unstructured student scripts.
- Research on handwritten text recognition shows that differences in handstrokes can reduce OCR accuracy. Variations in stroke pressure, shape, and sequence often create unclear character forms that are difficult to interpret. Jeon et al. (2026) found that even early handwriting exhibits noticeable differences in stroke formation that affect recognition. Similarly, Ram Kumar et al. (2023) noted that inconsistent stroke patterns make character segmentation and feature extraction more challenging.
- Errors in handwriting recognition directly affect automated grading accuracy. Gold and Zesch (2020) found that even minor OCR errors can alter the meaning of student responses, leading to incorrect scores, especially in essay-based assessments. Similarly, Karpagalakshmi et al. (2025) noted that while deep learning models perform well on structured data, their accuracy decreases with real-world handwriting variations.

The processing of handwritten responses introduces additional complexity. Ghadekar et al. (2025) integrated OCR with machine learning to automate grading of handwritten scripts, including mathematical expressions and diagrams. Similarly, Sumathi (2025) proposed a



dual-engine OCR framework combining TrOCR and EasyOCR to enhance transcription accuracy. Hiremath et al. (2024) further advanced this area by incorporating Large Language Models (LLMs) into multi-modal grading systems.

These studies collectively indicate that mobile-based capture systems must incorporate standardized templates and multi-engine OCR pipelines to ensure reliable digitization before AI grading can occur effectively.

Automated Grading Systems and Artificial Intelligence Applications

Automated grading systems have evolved from simple keyword-matching techniques to advanced semantic evaluation using transformer-based architectures and Large Language Models.

- Chaudhari (2025) introduced a Smart Paper Assessment System integrating OCR, NLP, and LLMs to digitize and semantically evaluate student responses. The study highlighted the inefficiencies and inconsistencies of manual grading while demonstrating the potential of AI to provide scalable and unbiased evaluation.
- Empirical findings by Himi et al. (2024) revealed that GPT-4 achieved error rates as low as 1.29% when compared with human graders, suggesting strong generalization capabilities even in zero-shot learning scenarios. However, Debelak and Ziegler (2024) emphasized the importance of ensuring fairness, validity, and psychometric reliability in AI-generated scores.



Other researchers have explored hybrid machine learning strategies. Hasan et al. (2023) combined clustering algorithms with optimization techniques to improve short-answer grading accuracy. Liu et al. (2022) and Gobrecht et al. (2024) applied deep learning models to capture semantic relationships beyond surface-level word matching.

Rubric-Based and Essay Scoring Frameworks

Rubric-based grading plays a critical role in maintaining pedagogical validity within automated systems. Explicit criteria provide measurable parameters that AI models can operationalize while preserving educational standards

- Safilian et al. (2025) developed RATAS, a rubric-aligned generative AI framework capable of producing both scores and detailed justifications. Their findings highlight the importance of explainability in automated grading systems.
- Research in Automated Essay Scoring (AES) further supports rubric integration. Santosh et al. (2022) demonstrated that machine learning models trained on pre-graded essays can replicate human scoring patterns. Drolia et al. (2017) applied regression-based NLP methods to evaluate grammar, coherence, vocabulary richness, and argument structure.
- Studies focusing on language diversity, such as those by Singh et al. (2023), Mahato (2022), and Rajagede (2021), emphasized the importance of contextual adaptability in AES systems. Meanwhile, Firoozi et al. (2023) and Gupta (2023) explored active learning and data augmentation techniques to improve grading performance with limited training data.



- These studies reinforce that rubric operationalization enhances grading transparency, reliability, and interpretability within AI-based evaluation frameworks.

Template-Based Standardization in Assessment

Standardized templates improve both OCR accuracy and grading consistency by constraining document layout variability. Elbourhamy (2024) demonstrated that structured exam formats enhance quality control and reduce inconsistencies in assessment.

- Chavan et al. (2023) and Sivaranjani et al. (2023) showed that semi-structured answer formats improve keyword extraction and semantic processing reliability. Ramani et al. (2024) further reported that standardized layouts allow automated systems to process hundreds of scripts efficiently while maintaining accuracy rates above 90%.
- However, Lim et al. (2021) cautioned that rigid standardization may limit legitimate variations in student responses. Therefore, template design must balance structural guidance with expressive flexibility.

System Architecture and Implementation Considerations

Effective implementation of automated grading systems requires scalable and user-centered system architecture.

- Lyu (2022) implemented a Browser/Server (B/S) model that separates the presentation layer from server-side processing, enabling concurrent user access and centralized system updates. Lin et al. (2024) introduced reliability detection mechanisms to flag anomalous



submissions for human review, enhancing trustworthiness in large-scale grading environments.

- Mobile capture integration requires preprocessing techniques such as perspective correction, lighting normalization, and noise reduction to ensure OCR effectiveness. Furthermore, Anjum et al. (2023) emphasized that AI-based grading systems must provide transparency and allow educator oversight to encourage adoption.
- Commercial assessment tools such as ZipGrade have demonstrated the practical viability of mobile-based optical scanning for objective examination evaluation. ZipGrade enables educators to grade multiple-choice bubble sheets using a smartphone camera, achieving rapid turnaround with minimal equipment. However, its functionality is restricted exclusively to pre-formatted bubble-sheet responses and does not support handwritten text, essay evaluation, or rubric-based semantic scoring (ZipGrade, 2023). The limitations of tools like ZipGrade reinforce the necessity of the proposed hybrid system, which extends mobile-capture grading capability to include all question types, particularly handwritten essays through the integration of transformer-based OCR and LLM-based semantic evaluation.

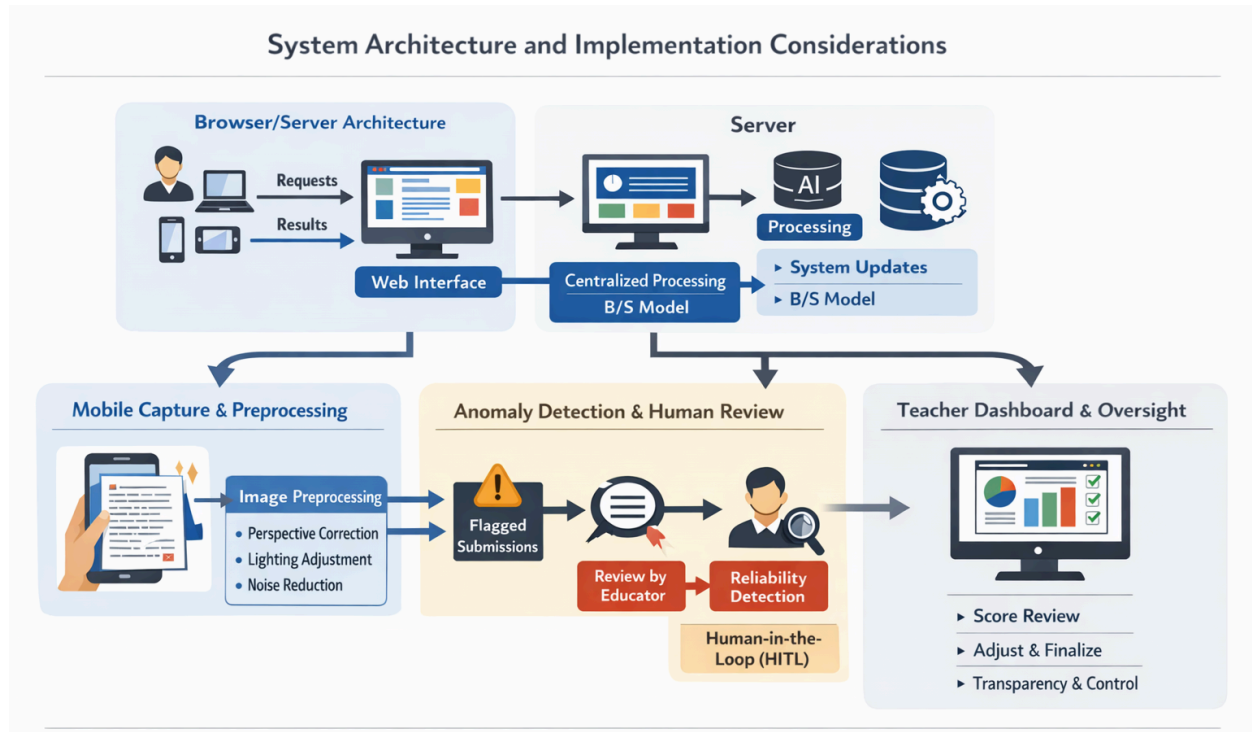


Figure 1. Proposed System Architecture for Automated Exam Scoring

Synthesis of Literature

The reviewed literature demonstrates that individual components of automated examination grading systems—OCR, AI semantic evaluation, rubric-based scoring, template standardization, and scalable architecture—have been extensively studied. However, limited research integrates all these components into a unified, end-to-end workflow that begins with physical paper capture and concludes with final grade validation.

The proposed AI-based hybrid automated examination paper checking system synthesizes these components into a cohesive framework. By combining mobile capture



technology, standardized templates, multi-engine OCR, rubric-aligned AI grading, and a web-based educator interface, the system bridges the gap between technical automation and pedagogical oversight.

Research Gap

Despite substantial advancements in automated grading research, several gaps remain. Few systems integrate mobile-based capture, structured templates, AI semantic grading, rubric management, and educator control within a single platform. Additionally, limited attention has been given to user-centered design and real-world adoption in resource-constrained educational environments.

The proposed system addresses these gaps by offering an end-to-end hybrid solution that enhances grading efficiency while preserving educator authority and assessment integrity.



CHAPTER 3 - METHODOLOGY

This chapter introduces the research design and methods applied in the design, development, and testing of the suggested Hybrid AI-Based Automated Examination Paper Checking System. It gives the research design, system development process, data collection methods, techniques of data analysis, and validation procedures, which were adopted in the study; a systematic approach to Design and Development Research (DDR) to ensure the system is accurate, reliable, efficient, and realistic in its applicability in academic grading.

Research Design

This study employs Design and Development Research (DDR) (Richey & Klein, 2007) as its primary research framework, supported by an Agile development methodology and a Mixed-Methods approach for system evaluation. DDR is appropriate for this study as it provides a structured process for the design, development, and evaluation of technology-based solutions through iterative refinement.

The study is composed of two major components: system development and system evaluation. The development phase follows Agile principles, allowing continuous improvement of the system based on testing and feedback. The evaluation phase utilizes a Mixed-Methods approach to comprehensively assess system performance and user experience.

The quantitative component measures system performance using objective metrics such as Character Error Rate (CER), Word Error Rate (WER), grading accuracy, inter-rater reliability,



and statistical comparison of pre-test and post-test perceived workload scores using the NASA Task Load Index (NASA-TLX).

The qualitative component captures educator perspectives through usability evaluation, focusing on system interaction, layout accessibility, trust, and adoption intent.

Within the evaluation phase, the study applies a within-subjects experimental design, where the same participants perform both traditional manual grading and GrAid-assisted grading. This design enables direct comparison of performance and workload while minimizing variability between participants.

This study will utilize four research instruments:

- 1. Performance Metrics Checklist:** A researcher-constructed checklist will be used to evaluate system accuracy through Character Error Rate (CER), Word Error Rate (WER), objective grading accuracy, and Pearson correlation between AI-generated and human-assigned essay scores.
- 2. NASA Task Load Index (NASA-TLX):** A standardized 6-dimension workload assessment will be administered before and after system testing to measure the professors' perceived mental effort and temporal demand.
- 3. System Usability Scale (SUS):** A standardized, pre-validated 10-item Likert instrument (Brooke, 1996) will be administered to five General Education professor evaluators to measure perceived usability following task completion, yielding a score out of 100%.



System Development Methodology

The development process follows an Agile-based Design and Development Research (DDR) framework, utilizing iterative cycles to refine the system through continuous testing and feedback. The process begins with the Requirement Analysis phase, where the manual grading pain points of General Education professors are analyzed to define the functional requirements for the mobile-responsive teacher dashboard and standardized A4 templates.

During the System Design and Development phase, the architecture is constructed in iterative sprints. This includes the implementation of a sophisticated hybrid OCR pipeline integrating Surya OCR for document layout and line detection, and Qwen2.5-VL-7B for vision-based handwritten text recognition. This phase also incorporates the Groq API (Llama 3.3-70B) to perform rubric-based semantic scoring and generate constructive feedback. This Agile approach allows for the real-time refinement of the "Human-in-the-Loop" workflow, ensuring that the flagging mechanism for low-confidence transcriptions is intuitive before entering the Implementation and Evaluation phase. In this final stage, system efficacy is validated by measuring time-on-task metrics and assessing user acceptability among selected faculty members through the NASA Task Load Index (NASA-TLX) and usability surveys.

Agile-based Design and Development Research (DDR)

Phase 1 - Requirement Analysis



Pain point analysis Grading inefficiencies identified by professors through consultation	Requirements definition Teacher dashboard Standardized templates System scope defined
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Iterative cycles feed back into Phase 1

Phase 2 – System Design & Development (Iterative Agile Cycles)	
Dual-engine OCR OMR– structured objective questions Qwen2.5-VL-7B-Instruct – complex handwritten essays	Semantic scoring Groq (Llama 3) AI-driven grading Automated feedback generation
Human-in-the-Loop Workflow Teacher review – Override & Correction – Real-time validation before deployment	

Refinement feedback feeds back into Phase 2

Phase 3 – Implementation & Evaluation	
Efficiency Testing Time-on task metrics measured & analyzed System performance benchmarked	Acceptability testing Professor satisfaction assessed & reported Usability reviewed for deployment

Figure: Agile-based DDR methodology showing three iterative phases with feedback loops.

Requirement Specification and Analysis



This section explains the capabilities of the GradeAID system from the user's perspective, bridging the physical handling of examination papers with advanced digital processing:

Statement of Valid Input

To process an examination successfully, the system requires the following inputs from the professor:

- **The Captured Exam:** A clear, high-resolution (Minimum **12 MP**) smartphone photograph of the student's physical test.
- **The File Format:** Images must be in standard **JPG** or **PNG** format and captured under readable lighting conditions.
- **The Template:** The examination must be written on the project's official, custom-downloadable paper template.
- **The Answer Key:** Instructor-defined reference answers and grading rubrics inputted directly via the web dashboard.

Detailed Specifications

- **Potential Users:** General Education professors and academic staff at Systems Plus College Foundation handling massive student volumes.
- **Speed and Response Time:** The backend inference server is expected to process a single photograph (executing edge detection, OCR, and Groq grading) within 20 to 40 seconds



CUDA-capable GPU server (NVIDIA RTX-class), comprising approximately 15–25 seconds for Qwen2.5-VL-7B-Instruct handwriting transcription and 3–8 seconds for Groq API semantic scoring.

- **Cost:** Minimal operating cost. By using pre trained open source models (SuryaOCR, Qwen2.5-VL-7B-Instruct, and Groq) running locally on a designated GPU server rather than paid cloud APIs (like OpenAI or GPT4), the school bypasses ongoing recurring token costs.
- **User-Friendliness:** The React.js frontend utilizes a clean, intuitive layout designed to replicate standard red-pen paper checking, lowering the learning curve for non-technical professors.
- **Performance:** Aims for a Character Error Rate (CER) $\leq 10\%$ and a reduction of at least 50% in the Temporal Demand and Effort sub-scales compared to traditional manual paper evaluation, as measured by a pre-test/post-test administration of the NASA Task Load Index.
- **Capacity:** The SQLite database is structured to support relational storage for standard class sizes (typically up to 30 students per automated batch).
- **Technology & Design Methods:** Leverages state-of-the-art Computer Vision (OpenCV) alongside deep learning Transformers (HuggingFace and Groq) built on a mobile-responsive Client-Server web architecture.

A. Functional Decomposition of the Project



The major functions required by the GrAid project design are broken down sequentially into five core processing modules.

Module	Sub-Function	Method / Tool	Description
1. Template & Capture	Physical Layout & Upload	React.js Web App, fpdf2, qrcode[pil]	Standardized templates are generated as PDFs with embedded QR codes and markers. Teachers capture them via smartphone and upload through the web interface.
2. Image Preprocessing	Rectification & Noise Removal	OpenCV 4.10+	Applies grayscale conversion, ArUco marker detection, perspective transformation (homography), and noise reduction to correct image distortions.

3. Hybrid Extraction	OCR Orchestration	Surya OCR, Qwen2.5-VL -7B-Instruct	Surya detects handwritten text regions and line segments, while Qwen2.5-VL transcribes handwritten responses into digital text.
4. AI Evaluation	Semantic Scoring	Groq API (Llama-3.3-70B-Versatile)	Evaluates student responses using rubric-based semantic analysis and generates both scores and feedback.
5. Validation Portal	Human-in-the-Loop (HITL)	Dashboard UI	Low-confidence scores are flagged. Professors can review, override, and manually adjust scores.

Table 1. Functional Decomposition of the Processing Pipeline

B. Selection of Design Criterion

Based on the operational parameters of the study, the following goals have been selected to measure the project's success:



- **Performance:** Achieve an objective grading accuracy of 95% and maintain a Pearson correlation coefficient (0.75%) matching AI-generated essay scores strictly to human evaluator standards.
- **Reliability:** The dual-engine OCR must maintain a strict threshold. Any complex handwriting that drops below standard confidence is actively flagged, shifting processing from the machine to the human evaluator to eliminate AI "hallucinations" or false grading.
- **Societal Factors (Human Interface):** The primary environment is a physical-to-digital bridge. The human interface must alleviate teacher burnout and fatigue while accelerating the speed of delayed academic feedback to students without replacing human educators' final judgment.
- **Environmental Factors / Materials:** By utilizing digital uploads on regular standard office paper templates rather than requiring expensive specialized OMR hardware machines or heavy plastic scan sheets, the system promotes cost-effective resource management.

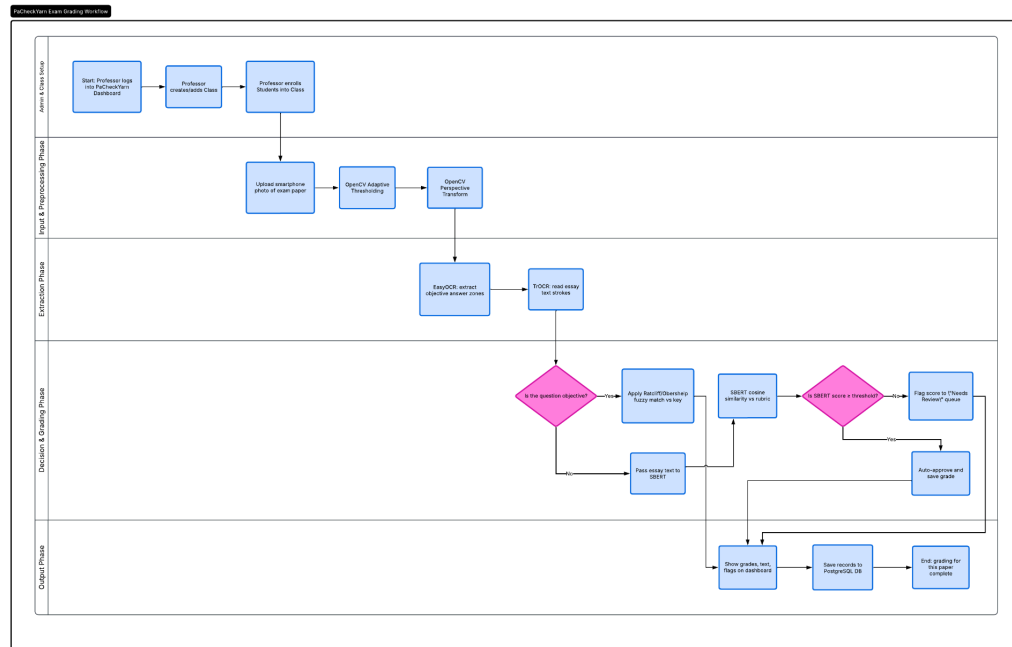


Figure 1. System Flowchart

System Design Specification

The architecture of GrAId is built upon a modular integration of specialized tools and frameworks designed to balance grading accuracy with practical usability:

- **React.js (Frontend):** Serves as the primary interface layer for the teacher dashboard, managing image uploads, rubric input, score display, and professor overrides in a mobile-responsive web environment.
- **Python / FastAPI (Backend):** Handles all server-side processing, including image preprocessing, OCR orchestration, scoring computation, confidence thresholding, and



API endpoint management. Uvicorn is used as the ASGI server, while Pydantic ensures structured data validation..

- OpenCV: Utilized in the image preprocessing and scan alignment pipeline to perform grayscale conversion, noise reduction, ArUco marker detection, and perspective transformation (homography), ensuring accurate cropping and alignment of captured examination sheets.
- Surya OCR + Qwen2.5-VL-7B-Instruct: A hybrid vision-based OCR pipeline where Surya is used for handwritten line and region detection, and Qwen2.5-VL-7B-Instruct performs vision-language transcription of handwritten responses into digital text using transformer-based inference.
- Groq API (Llama-3.3-70B-Versatile): Serves as the semantic scoring engine for essay-type responses. It evaluates student answers against instructor-defined rubric criteria and generates both numerical scores and detailed feedback, enabling explainable AI-assisted grading.
- SQLite with SQLAlchemy: Used as the lightweight relational database for storing examination records, transcribed responses, rubric entries, AI-generated scores, confidence values, and professor override decisions. Alembic is utilized for database schema migrations.

Logical Specification



This section will define and organize information requirements and logic of the proposed automated system of checking test papers. The system consists of various modules, which are interconnected and which process image-input, text-extraction, answers evaluation, and presentation of results. The process flow of these modules is illustrated in *Figure 1*, which shows the overall processing pipeline of the system.

The system takes as input digital image of answer sheets which includes JPEG, PNG and WEBP formats, but with a recommended minimum resolution of around 12 megapixels to be fed in the system to enable proper processing. Images uploaded are validated and preprocessed such as resizing, grayscale conversion, deskewing, denoising, and adaptive thresholding to enhance readability in extracting text.

Its processing logic is a hybrid Optical Character Recognition (OCR) pipeline, where text regions are initially located and interpreted through transformer-based models. Text that has been extracted is evaluated using several assessment techniques, such as analyzing pattern matching (objective questions), fuzzy matching of strings (identification-type questions), and AI scoring (essays). The system uses pre-established rules and scoring criteria to come up with precise and comparable results.

The output is formatted into three formats namely; an interactive web-based dashboard with per-question marks and feedback, a JSON-based responding API with detailed evaluation marks, and stored data in a central database. Teachers have override capability to allow them to

make changes on the scores manually, making sure that they have human validation where required. The system has a base processing time range of 15-30 seconds per paper on a CPU based environment with a 20MB maximum file size per image on batch upload. Future development iterations will incorporate data encryption and role-based access controls to meet institutional data privacy requirements.



Figure 2. Functional Decomposition of the Processing Pipeline



Physical Specification

These are the following hardware specifications in order to download the necessary software and technologies that are compatible to run the application. The physical specification is shown in the following tables below.

Category	Component	Minimum Requirements	Suggested Requirements
Mobile Device	Camera	12 MP with Autofocus	48 MP+ with Ultra-Wide/Macro
	Connectivity	4G LTE / Wi-Fi	5G / Wi-Fi 6
	Browser	Chrome/Safari (Latest)	Chrome/Safari (Latest)

Table 1. Mobile Specifications

Category	Component	Minimum Requirements	Suggested Requirements
Processing Server/PC	CPU	Intel Core i5 (11th Gen)	Intel Core i7 (13th Gen) or higher
	RAM	8 GB DDR4	16 GB - 32 GB DDR5



	Storage	256 GB SSD	512 GB NVMe SSD
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Table 2. PC Specifications

Category	Component	Minimum Requirements	Suggested Requirements
Network	Bandwidth	10 Mbps	50 Mbps+

Table 3. Network Specifications

Implementation

The implementation of the GrAid system centers on a robust FastAPI and React 18 architecture, utilizing Python 3.11 to orchestrate a sophisticated AI pipeline. The physical-to-digital transition begins with custom A4 templates generated via fpdf2, which feature ArUco markers and QR codes for automated metadata extraction and perspective correction using OpenCV. Once aligned, the system employs a hybrid OCR strategy, combining Surya for structural line detection and a 4-bit quantized Qwen2.5-VL-7B model for high-fidelity handwriting transcription.

Subjective evaluation is handled by the Llama-3.3-70B model via the Groq API, which provides rubric-aligned semantic scoring and feedback. To ensure reliability, a Human-in-the-Loop mechanism flags low-confidence results for manual review through a mobile-responsive Tailwind CSS dashboard. The entire infrastructure is deployed using



Cloudflare Quick Tunnels, providing secure, public HTTPS access for real-world testing across diverse handwriting styles and lighting conditions.

Testing and Evaluation

The testing and evaluation phase of this study will employ a two-session, within-subjects design involving five General Education professors from Systems Plus College Foundation, who will serve as both the inter-rater reliability raters and the user acceptance testing evaluators. This dual-role design will be structured sequentially to prevent contamination: all five professors will first complete a traditional manual grading session (Session 1), which will establish the human ground truth for essay scoring and the pre-test NASA-TLX baseline, before proceeding to a GrAid-assisted grading session (Session 2), which will generate the post-test NASA-TLX scores and all usability evaluation data. The selection of a five-evaluator sample is consistent with the foundational guidelines for usability inspection established by Nielsen (1994) and is supported by recent literature on expert-driven formative evaluation (Tawfik et al., 2022; Schmidt & Huang, 2024). Within the Design and Development Research (DDR) framework (Richey & Klein, 2007), this sample size is appropriate for the Evaluation Phase, as it focuses on identifying critical system-user interaction patterns and validating the tool's functionality among a homogeneous group of subject matter experts. By involving five General Education professors, the study reaches a point of data saturation where major usability obstacles are identified and the system's impact on cognitive load can be empirically assessed. This rigorous evaluation serves as



the internal validation of the GrAid system, providing a robust empirical basis for its effectiveness in professional grading workflows.

Evaluator Configuration and Session Design

Five General Education professors will be recruited from Systems Plus College Foundation to participate in both evaluation sessions. To preserve the integrity of the human ground truth, all professors will complete Session 1, the traditional manual grading session, before any exposure to GrAid. During Session 1, each professor will independently grade the same set of student essay responses using the instructor-defined rubric provided by the research team, with no interaction or discussion permitted between raters. Their individual scores will be recorded and submitted to inter-rater reliability analysis prior to any comparison with GrAid's output. Immediately following the completion of Session 1, professors will respond to the NASA Task Load Index (NASA-TLX) to capture their perceived workload under traditional manual grading conditions, establishing the pre-test baseline.

Session 2 will be conducted after a minimum rest interval of one week to reduce fatigue and recency bias. During Session 2, the same five professors will use GrAid to grade an equivalent set of student examination papers through the teacher dashboard, uploading photographs, reviewing AI-generated transcriptions, inspecting rubric-based essay scores and justifications, and exercising override capabilities where necessary. Upon completion, professors



will again respond to the NASA-TLX, yielding the post-test workload scores. They will subsequently be administered the System Usability Scale (SUS).

Inter-Rater Reliability

Inter-rater reliability among the five professor raters will be assessed using Fleiss' Kappa (Fleiss, 1971), which is the standard extension of Cohen's Kappa designed specifically for three or more raters assigning categorical or ordinal ratings to the same set of subjects (McHugh, 2012; Landis & Koch, 1977). Unlike Cohen's Kappa, which is limited to exactly two raters, Fleiss' Kappa accommodates any number of raters and is therefore the appropriate statistic for this study's five-rater configuration. Each of the five professors will independently assign numerical scores to the same set of student essay responses using the instructor-defined rubric, and Fleiss' Kappa will be computed across all five raters' scores to determine the degree of agreement beyond what would be expected by chance.

A minimum Fleiss' Kappa of $\kappa \geq 0.70$, corresponding to the "substantial agreement" range as defined by Landis and Koch (1977), will be required to confirm that the human-assigned scores used as ground truth reflect consistent and defensible professional judgment. This threshold was selected over the more lenient $\kappa \geq 0.61$ minimum used in some prior studies to strengthen the validity of the ground truth baseline, particularly given the subjectivity inherent in essay evaluation. Once inter-rater reliability is confirmed, the mean of the five professors' scores will serve as the verified ground truth for all subsequent comparisons with



GrAid's AI-generated output. In instances where Fleiss' Kappa falls below $\kappa = 0.70$, the research team will facilitate a structured consensus discussion among the five raters to resolve systematic disagreements in rubric interpretation prior to finalizing the ground truth scores.

OCR Accuracy Assessment

OCR accuracy will be assessed using Character Error Rate (CER) and Word Error Rate (WER) computed via the JiWER Python library, comparing the hybrid OCR pipeline's output, produced by Surya for line detection followed by Qwen2.5-VL-7B-Instruct for transcription, against verified ground truth transcriptions independently produced by two members of the research team and cross-validated against the standardized answer sheet template. CER and WER will apply primarily to Identification and Essay question types, as Multiple Choice and True/False responses will be processed through exact-match and OMR pathways that do not depend on OCR text accuracy.

This study adopts the performance benchmarks reported by J. Kumar and Dharshan (2025), who achieved a CER of 4.57% and WER of 12.3% using a comparable deep learning OCR pipeline on the IAM dataset under similarly varied real-world conditions. The study will set target thresholds of $\text{CER} \leq 4.57\%$ and $\text{WER} \leq 12.3\%$, with minimum acceptable thresholds of $\text{CER} \leq 10\%$ and $\text{WER} \leq 15\%$ respectively. In cases where OCR output falls below the system's confidence threshold, the affected response will be automatically flagged for mandatory



human review through the teacher dashboard, serving as the system's primary safety net against transcription-related grading errors.

Grading Accuracy Assessment

Grading accuracy will be evaluated separately by question type to reflect the distinct mechanisms of the hybrid grading engine. For Multiple Choice and True/False questions, accuracy will be measured by comparing GrAid's exact-match output against a verified answer key, targeting a correctness rate of $\geq 95\%$, computed as percentage accuracy. For Identification questions, accuracy will likewise be measured against the verified answer key using fuzzy string matching, with full credit defined as responses achieving a similarity score of $\geq 85\%$ against the reference answer. Partial credit scoring, which applies to responses with similarity scores between $\geq 60\%$ and $< 85\%$, is implemented in the system but is outside the scope of the current evaluation due to the absence of a standardized partial credit rubric for the test items used. The target accuracy for Identification full credit grading is also $\geq 95\%$.

For essay responses, evaluation will follow a three-part process. First, Fleiss' Kappa will be computed across the five independent professor raters as described in the inter-rater reliability section above. Once a minimum $\kappa \geq 0.70$ is confirmed, the mean of the five professors' scores will serve as the verified ground truth. GrAid's AI-generated essay scores, produced by the Groq API using the Llama-3.3-70B-Versatile model, will then be compared against this ground truth using Pearson correlation coefficient, targeting $r \geq 0.75$, following the approach established by



Santosh et al. (2022). Finally, rubric alignment will be assessed by asking the same five professor evaluators to rate the quality of Groq's written justification for each essay score on a 5-point Likert scale, where 1 indicates the justification does not reflect the rubric criteria at all and 5 indicates it fully and accurately reflects the rubric criteria. A mean score of $\geq 4.0/5.0$ will be set as the target to confirm that GrAid's essay scoring is not only numerically consistent with human judgment but also pedagogically explainable.

Confidence Threshold and Human-in-the-Loop Flagging

The reliability of the confidence threshold and Human-in-the-Loop flagging mechanism will be evaluated using Flagging Precision and Flagging Recall. Flagging Precision will measure the proportion of flagged responses that were confirmed to contain actual grading errors upon professor review, targeting $\geq 80\%$. Flagging Recall will measure the proportion of all actual errors in the dataset that were successfully detected and flagged by the system, targeting $\geq 70\%$. Both metrics will be computed by cross-referencing the system's flagging log, recorded in GrAid's SQLite database at the time of professor review, against the verified answer key. Every instance where a professor reviews, adjusts, or overrides an AI-generated score will be captured in this log, enabling post-evaluation analysis of flagging performance.

Perceived Task Load Assessment (NASA-TLX)

The NASA Task Load Index (NASA-TLX) (Hart & Staveland, 1988) will be administered to the five General Education professor evaluators before and after system use to



measure perceived task load reduction. The NASA-TLX is a standardized, widely validated six-dimension workload assessment instrument covering mental demand, physical demand, temporal demand, performance, effort, and frustration, and has been cited in over 4,400 human factors studies since its development (Hart, 2006). Recent research by Bhadila et al. (2026) reinforces the instrument's utility in modern educational settings, demonstrating that AI assistance can lead to a significant reduction in perceived task load (median scores of 21.7 vs. 41.7) when compared to manual baseline tasks. The instrument is explicitly designed to support repeated administration across different task conditions experienced by the same participants, making it appropriate for the pre/post within-subjects comparison that will be employed in this study (Hart & Staveland, 1988).

Professors will first complete Session 1, the traditional manual grading session, and will respond to the NASA-TLX immediately after completion to establish the pre-test baseline workload score. Following the minimum one-week interval, they will complete Session 2 using GrAid and will respond to the NASA-TLX again as the post-test measure. The overall composite NASA-TLX score will be used as the primary outcome measure, targeting a reduction of $\geq 50\%$ between pre-test and post-test administrations, analyzed using a paired-sample t-test. The within-subjects sequential design was selected over manual timing because individual grading habits, preferences, and natural speed variations among professors would introduce significant measurement bias in a time-on-task comparison, particularly given the pilot-scale sample size of five evaluators.



User Acceptance Testing

User Acceptance Testing (UAT) will be conducted with the same five General Education professors following the completion of Session 2. The five-evaluator sample is justified by Nielsen's (1994) established guideline that five evaluators are sufficient to identify the majority of usability problems in a system interface, a standard reaffirmed by recent research as the optimal threshold for expert-driven formative evaluation. Following the GrAid-assisted grading session, professors will be administered the System Usability Scale (SUS) (Brooke, 1996), a standardized 10-item Likert instrument. The continued use of the SUS as the industry standard for educational technology is supported by Aksu et al. (2025), who emphasize its robustness in measuring "perceived ease of use" alongside cognitive load metrics. The instrument yields a composite score out of 100, with a target of ≥ 70 to meet acceptable usability standards. To complement this, a structured observation and interview guide will capture qualitative feedback on usability barriers, system strengths, and adoption intent, constructs which Bhadila et al. (2026) identified as critical predictors of AI integration in professional settings. Responses will be analyzed via thematic analysis as the qualitative component of the study's mixed-methods approach.

Metric	Target	Method	Statistical Treatment
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OCR Character Error Rate (CER)	Target: $\leq 4.57\%$ / Minimum: $\leq 10\%$	Ground-truth comparison via JiWER Python library	Descriptive statistics
OCR Word Error Rate (WER)	Target: $\leq 12.3\%$ / Minimum: $\leq 15\%$	Ground-truth comparison via JiWER Python library	Descriptive statistics
MC/TF Grading Accuracy	$\geq 95\%$	Answer key comparison (exact-match)	Percentage accuracy
Identification Grading Accuracy	$\geq 95\%$	Answer key comparison (fuzzy-match, full credit only $\geq 90\%$ similarity)	Percentage accuracy
Inter-Rater Reliability (Fleiss' Kappa)	$\kappa \geq 0.70$	Five independent professor raters grading the same essay set	Fleiss' Kappa
Groq-Human Score Correlation	$r \geq 0.75$	Averaged scores from five independent	Pearson correlation



		professor raters as ground truth	
Rubric Alignment	Mean $\geq 4.0/5.0$	Professors rate Groq's written justification against rubric criteria using a 5-point Likert scale	Mean score (Likert)
Flagging Precision	$\geq 80\%$	Cross-reference flagged responses against verified answer key to confirm actual errors	Percentage accuracy
Flagging Recall	$\geq 70\%$	Cross-reference all actual errors against flagged responses to confirm detection rate	Percentage accuracy
Trust and Transparency Scale	Mean $\geq 4.0/5.0$	Professors rate confidence in flagging system and dashboard transparency using a 5-item Likert scale	Mean score (Likert)



Perceived Task Load Reduction (NASA-TLX Overall Score)	$\geq 50\%$ reduction in composite score	Pre/post NASA-TLX administered before traditional manual grading (Session 1) and after GrAid-assisted grading (Session 2)	Paired-sample t-test
System Usability (SUS Score)	≥ 70	Professor survey administered after GrAid-assisted grading session	Mean score (Likert)

Table 4. System Evaluation Results



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